

1 Summary

Model (Run-name): model3 (test)

Iteration: 12

TAC: 63966.988672050866

Took: 929.5291128158569s

Total Time: 1863.137009859085s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

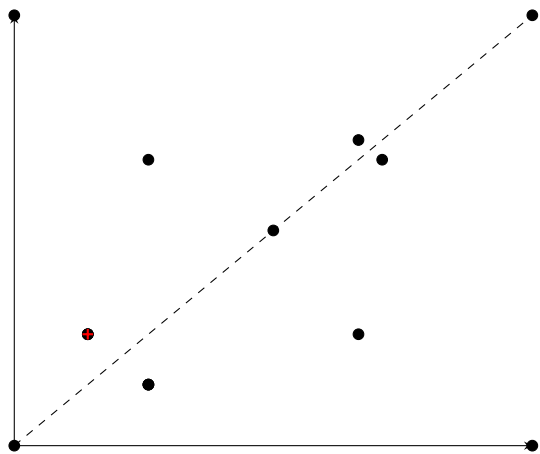
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	1.0	$dt_{3,5,1}$	22.3	$q_{3,5,1}$	1544.2900000000002
$z_{2,1,1}$	1.0	$dt_{3,5,2}$	29.056081081081047	$q_{4,3,2}$	1539.72
$z_{2,2,2}$	1.0	$dt_{4,3,2}$	50.0	$q_{5,4,2}$	1634.85
$z_{3,5,1}$	1.0	$dt_{4,3,3}$	111.1	$q_{cu,2}$	357.4100000000112
$z_{4,3,2}$	1.0	$dt_{5,4,2}$	22.2	$q_{cu,3}$	839.9899999999994
$z_{5,4,2}$	1.0	$dt_{5,4,3}$	24.335593220337053	$q_{cu,5}$	724.5599999999989
$z_{cu,2}$	1.0	$dt_{cu,2}$	131.51792452830296	$t_{1,1}^{(H)}$	160.0
$z_{cu,3}$	1.0	$dt_{cu,3}$	82.35608108108103	$t_{1,2}^{(H)}$	160.0
$z_{cu,5}$	1.0	$dt_{cu,5}$	66.53559322033841	$t_{1,3}^{(H)}$	93.3
$A_{1,1,2}^{\beta}$	6.886616396612075	$f_{1,1,2}^H$	8.8	$t_{2,1}^{(H)}$	248.9
$A_{2,1,1}^{\beta}$	1.550352329714329	$f_{2,1,1}^H$	1.557515751575105	$t_{2,2}^{(H)}$	232.57547169811426
$A_{2,2,2}^{\beta}$	6.318237400857532	$f_{2,2,2}^H$	6.828876590153504	$t_{2,3}^{(H)}$	171.51792452830296
$A_{3,5,1}^{\beta}$	12.923002798326602	$f_{3,5,1}^H$	9.585909373060213	$t_{3,1}^{(H)}$	226.7
$A_{4,3,2}^{\beta}$	6.675846741185239	$f_{4,3,2}^H$	12.6	$t_{3,2}^{(H)}$	122.35608108108103
$A_{5,4,2}^{\beta}$	14.143774763563213	$f_{5,4,2}^H$	12.264441110277566	$t_{3,3}^{(H)}$	122.35608108108103
$A_{cu,2}^{\beta}$	0.6686318174746403	$f_{1,1,2}^C$	6.60185172435294	$t_{4,1}^{(H)}$	271.1
$A_{cu,3}^{\beta}$	5.422793392351579	$f_{2,1,1}^C$	7.6	$t_{4,2}^{(H)}$	271.1
$A_{cu,5}^{\beta}$	5.324365401359005	$f_{2,2,2}^C$	6.1	$t_{4,3}^{(H)}$	148.9
$A_{1,1,2}$	24.927400386350968	$f_{3,5,1}^C$	13.9	$t_{5,1}^{(H)}$	198.9
$A_{2,1,1}$	2.1978599029887658	$f_{4,3,2}^C$	8.4	$t_{5,2}^{(H)}$	198.9
$A_{2,2,2}$	22.02028770105733	$f_{5,4,2}^C$	17.3	$t_{5,3}^{(H)}$	106.53559322033841
$A_{3,5,1}$	71.16597394333074	$RecLMTD_{1,1,2}$	0.03609835479146401	$t_{1,1}^{(C)}$	160.0
$A_{4,3,2}$	23.668855770059167	$RecLMTD_{2,1,1}$	0.010796237387543497	$t_{1,2}^{(C)}$	137.23157894736988
$A_{5,4,2}$	82.71968770371944	$RecLMTD_{2,2,2}$	0.02891989392299058	$t_{1,3}^{(C)}$	60.0
$A_{cu,2}$	2.696334685145226	$RecLMTD_{3,5,1}$	0.039170802020236484	$t_{2,1}^{(C)}$	221.7
$A_{cu,3}$	16.738084386188994	$RecLMTD_{4,3,2}$	0.013066354534948102	$t_{2,2}^{(C)}$	221.7
$A_{cu,5}$	16.234975184591587	$RecLMTD_{5,4,2}$	0.043008064683710046	$t_{2,3}^{(C)}$	115.6
$dt_{1,1,2}$	22.768421052630117	$RecLMTD_{cu,2}$	0.009423141115769086	$t_{3,1}^{(C)}$	221.1
$dt_{1,1,3}$	33.3	$RecLMTD_{cu,3}$	0.016937548933035187	$t_{3,2}^{(C)}$	221.1
$dt_{2,1,1}$	88.9	$RecLMTD_{cu,5}$	0.01904566958709165	$t_{3,3}^{(C)}$	37.8
$dt_{2,1,2}$	95.34389275074437	$q_{1,1,2}$	586.96	$t_{4,1}^{(C)}$	176.7
$dt_{2,2,2}$	10.875471698114254	$q_{2,1,1}$	173.039999999998905	$t_{4,2}^{(C)}$	176.7
$dt_{2,2,3}$	55.91792452830191	$q_{2,2,2}$	647.2099999999999	$t_{4,3}^{(C)}$	82.2

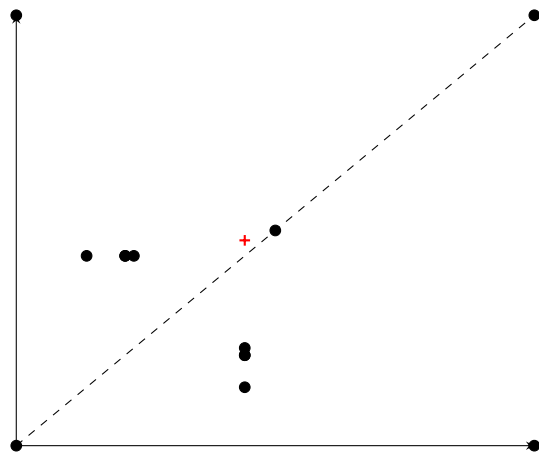
Variable	Value	Variable	Value
$t_{5,1}^{(C)}$	204.4	$b_{1,1,2}^{C,out}$	983.0711034611764
$t_{5,2}^{(C)}$	93.3	$b_{2,1,1}^{C,out}$	1216.0
$t_{5,3}^{(C)}$	93.3	$b_{2,2,2}^{C,out}$	1352.37
$t_{1,1,2}^H$	93.3	$b_{3,5,1}^{C,out}$	2841.1600000000003
$t_{2,1,1}^H$	137.8	$b_{4,3,2}^{C,out}$	1857.24
$t_{2,2,2}^H$	137.8	$b_{5,4,2}^{C,out}$	3056.91
$t_{3,5,1}^H$	65.6		
$t_{4,3,2}^H$	148.9		
$t_{5,4,2}^H$	65.6		
$t_{1,1,2}^C$	148.90838881383138		
$t_{2,1,1}^C$	160.0		
$t_{2,2,2}^C$	221.7		
$t_{3,5,1}^C$	204.4		
$t_{4,3,2}^C$	221.1		
$t_{5,4,2}^C$	176.7		
$b_{1,1,2}^{H,in}$	1408.0		
$b_{2,1,1}^{H,in}$	387.66567056704366		
$b_{2,2,2}^{H,in}$	1588.2291941231406		
$b_{3,5,1}^{H,in}$	2173.12565487275		
$b_{4,3,2}^{H,in}$	3415.8599999999997		
$b_{5,4,2}^{H,in}$	2439.3973368342085		
$b_{1,1,2}^{H,out}$	821.0399999999989		
$b_{2,1,1}^{H,out}$	214.6256705670546		
$b_{2,2,2}^{H,out}$	941.0191941231407		
$b_{3,5,1}^{H,out}$	628.8356548727501		
$b_{4,3,2}^{H,out}$	1876.1399999999996		
$b_{5,4,2}^{H,out}$	804.5473368342084		
$b_{1,1,2}^{C,in}$	396.1111034611764		
$b_{2,1,1}^{C,in}$	1042.9599999999818		
$b_{2,2,2}^{C,in}$	705.16		
$b_{3,5,1}^{C,in}$	1296.87		
$b_{4,3,2}^{C,in}$	317.52		
$b_{5,4,2}^{C,in}$	1422.0600000000002		

3 *RecLMTD* Tangent Points

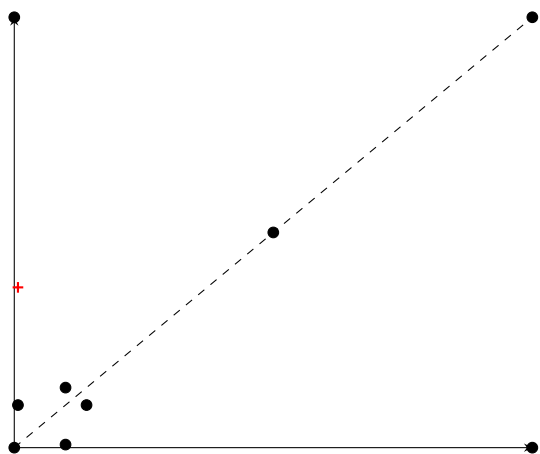
- (1, 1, 2)
 - Value: 0.036098
 - Error: 0.000000967497611
 - Relative error: 0.000026800990983
- (2, 1, 1)
 - Value: 0.010796
 - Error: 0.000063368649462
 - Relative error: 0.005835262278018
- (2, 2, 2)
 - Value: 0.028920
 - Error: 0.007431923723805
 - Relative error: 0.204444349826346
- (3, 5, 1)
 - Value: 0.039171
 - Error: 0.0000000000000000
 - Relative error: -0.0000000000000003
- (4, 3, 2)
 - Value: 0.013066
 - Error: 0.000000874453881
 - Relative error: 0.000066919610992
- (5, 4, 2)
 - Value: 0.043008
 - Error: 0.0000000000000001
 - Relative error: 0.0000000000000017
- $cu, 2$
 - Value: 0.009423
 - Error: 0.001221013191136
 - Relative error: -0.148865416676716
- $cu, 3$
 - Value: 0.016938
 - Error: 0.000000923579564
 - Relative error: 0.000054525551996
- $cu, 5$
 - Value: 0.019046
 - Error: 0.000000317526668
 - Relative error: 0.000016671578435



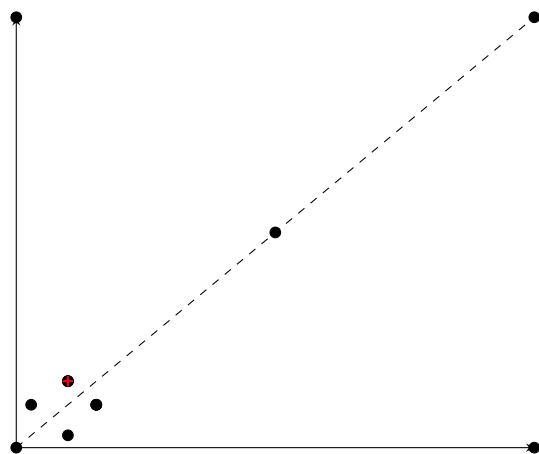
(a) *RecLMTD* - (1, 1, 2)



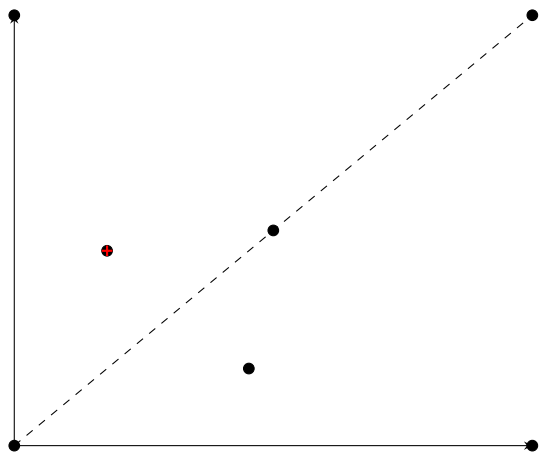
(b) *RecLMTD* - (2, 1, 1)



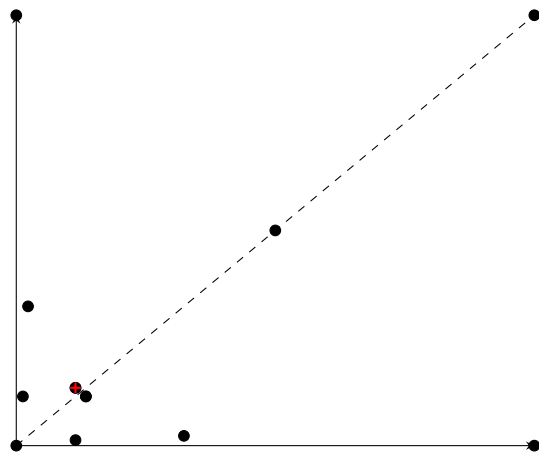
(c) *RecLMTD* - (2, 2, 2)



(d) *RecLMTD* - (3, 5, 1)



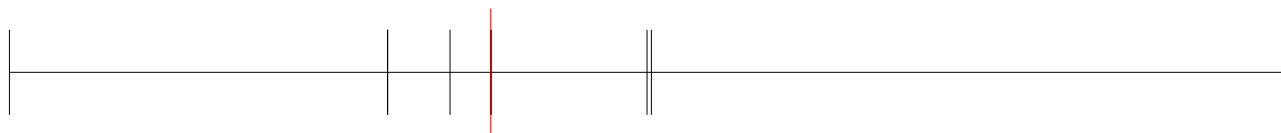
(e) *RecLMTD* - (4, 3, 2)



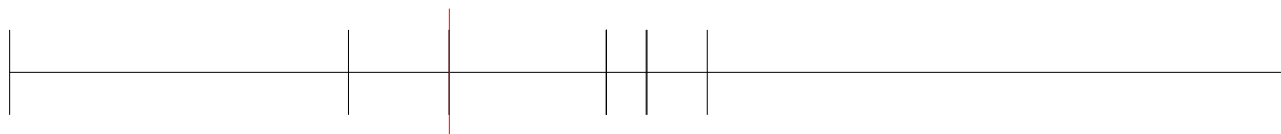
(f) *RecLMTD* - (5, 4, 2)



(a) *RecLMTD* - $cu, 2$



(b) *RecLMTD* - $cu, 3$



(c) *RecLMTD* - $cu, 5$

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 2)
 - Value: 1408.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 387.66567056704366
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 2)
 - Value: 1588.2291941231406
 - Absolute error: 2.0691004465334117e-11
 - Relative error: -1.3027719514221199e-14
- (3, 5, 1)
 - Value: 2173.12565487275
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 3415.8599999999997
 - Absolute error: 4.547473508864641e-13
 - Relative error: -1.3312821687260722e-16
- (5, 4, 2)
 - Value: 2439.3973368342085
 - Absolute error: 4.547473508864641e-13
 - Relative error: 1.8641790905479319e-16

4.2 $b^{H,\text{out}}$

- (1, 1, 2)
 - Value: 821.0399999999989
 - Absolute error: 1.1027623258996755e-11
 - Relative error: -1.3431286245489566e-14
- (2, 1, 1)
 - Value: 214.6256705670546
 - Absolute error: 5.115907697472721e-12
 - Relative error: 2.3836420331064273e-14
- (2, 2, 2)

- Value: 941.0191941231407
 - Absolute error: 1.2278178473934531e-11
 - Relative error: -1.3047744988215047e-14
- (3, 5, 1)
 - Value: 628.8356548727501
 - Absolute error: 1.1368683772161603e-13
 - Relative error: 1.8078942700000288e-16
- (4, 3, 2)
 - Value: 1876.1399999999996
 - Absolute error: 4.547473508864641e-13
 - Relative error: -2.4238455066597593e-16
- (5, 4, 2)
 - Value: 804.5473368342084
 - Absolute error: 1.1368683772161603e-13
 - Relative error: 1.4130534341081695e-16

4.3 $b^{C,in}$

- (1, 1, 2)
 - Value: 396.1111034611764
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 1042.95999999999818
 - Absolute error: 2.9103830456733704e-11
 - Relative error: -2.7905030352778054e-14
- (2, 2, 2)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 1296.87
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0

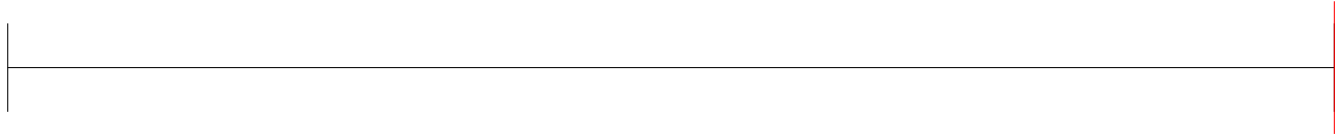
- (5, 4, 2)
 - Value: 1422.0600000000002
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

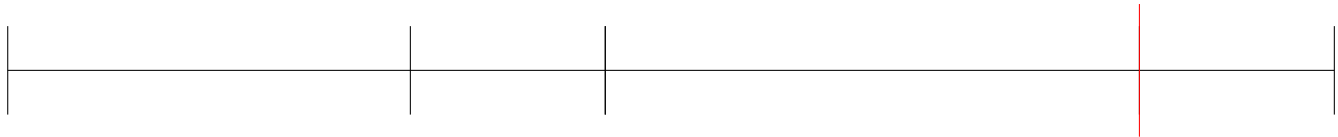
- (1, 1, 2)
 - Value: 983.0711034611764
 - Absolute error: 3.433342499192804e-11
 - Relative error: -3.4924660964040575e-14
- (2, 1, 1)
 - Value: 1216.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 2)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 2841.1600000000003
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 1857.24
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 2)
 - Value: 3056.91
 - Absolute error: 0.0
 - Relative error: 0.0



(a) $t^{(H)} - (1, 2)$



(a) $t^{(H)} - (2, 1)$



(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (3, 1)$



(a) $t^{(H)} - (4, 2)$



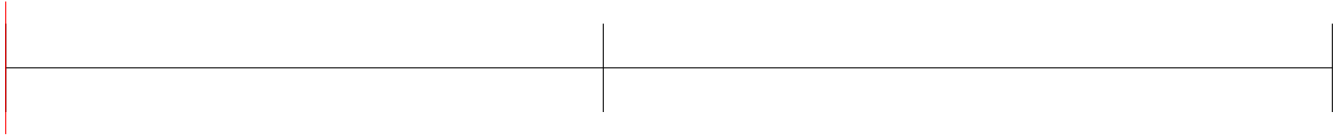
(a) $t^{(H)} - (5, 2)$



(a) $t^{(H)} - (1, 1, 2)$



(a) $t^{(H)} - (2, 1, 1)$



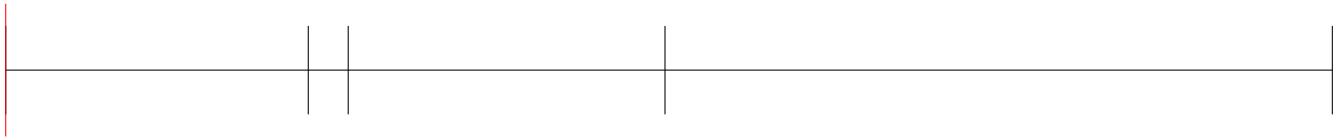
(a) $t^{(H)} - (2, 2, 2)$



(a) $t^{(H)} - (3, 5, 1)$



(a) $t^{(H)} - (4, 3, 2)$



(a) $t^{(H)} - (5, 4, 2)$



(a) $t^{(C)} - (1, 3)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



(a) $t^{(C)} - (5, 2)$



$$(a) \ t^{(C)} - (3, 3)$$



$$(a) \ t^{(C)} - (4, 3)$$



$$(a) \ t^{(C)} - (1, 1, 2)$$



$$(a) \ t^{(C)} - (2, 1, 1)$$



$$(a) \ t^{(C)} - (2, 2, 2)$$



$$(a) \ t^{(C)} - (3, 5, 1)$$



$$(a) \ t^{(C)} - (4, 3, 2)$$



$$(a) \ t^{(C)} - (5, 4, 2)$$

5 Area Breakpoints

5.1 A

- $(1, 1, 2)$
 - Value: 24.927400
 - Error: 0.000000000000711
 - Relative error: -0.000000000000029
- $(2, 1, 1)$
 - Value: 2.197860
 - Error: 0.000000000000050
 - Relative error: -0.000000000000023
- $(2, 2, 2)$
 - Value: 22.020288
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000
- $(3, 5, 1)$
 - Value: 71.165974
 - Error: 0.000000000000142
 - Relative error: -0.000000000000002
- $(4, 3, 2)$
 - Value: 23.668856
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000
- $(5, 4, 2)$
 - Value: 82.719688
 - Error: 0.000000000002160
 - Relative error: 0.000000000000026

5.2 A_{CU}

- $cu, 2$
 - Value: 2.696335
 - Error: 1.265929863310227
 - Relative error: 0.319496552496401
- $cu, 3$
 - Value: 16.738084
 - Error: 0.000000000000504
 - Relative error: -0.000000000000030
- $cu, 5$

- Value: 16.234975
- Error: 0.000001704847161
- Relative error: 0.000000105010754



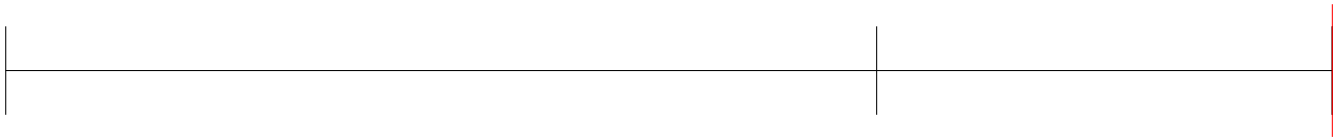
(a) $A^\beta - (1, 1, 2)$



(a) $A^\beta - (2, 1, 1)$



(a) $A^\beta - (2, 2, 2)$



(a) $A^\beta - (3, 5, 1)$



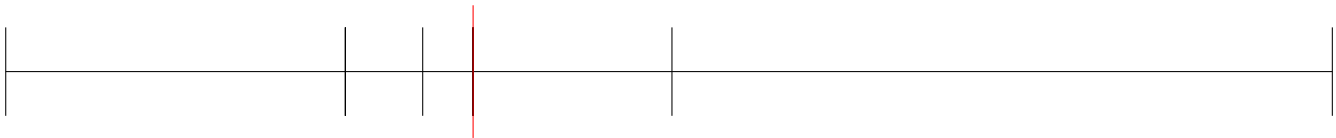
(a) $A^\beta - (4, 3, 2)$



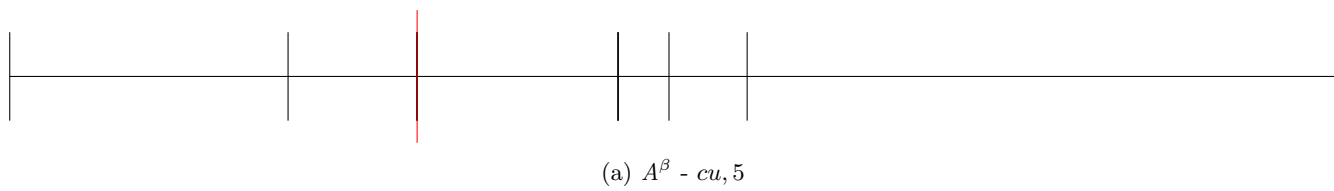
(a) $A^\beta - (5, 4, 2)$



(a) $A^\beta - cu, 2$



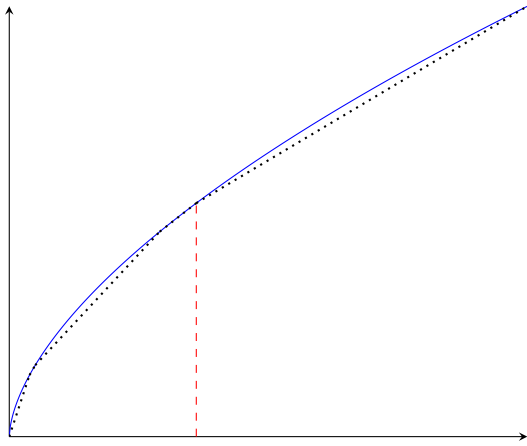
(a) $A^\beta - cu, 3$



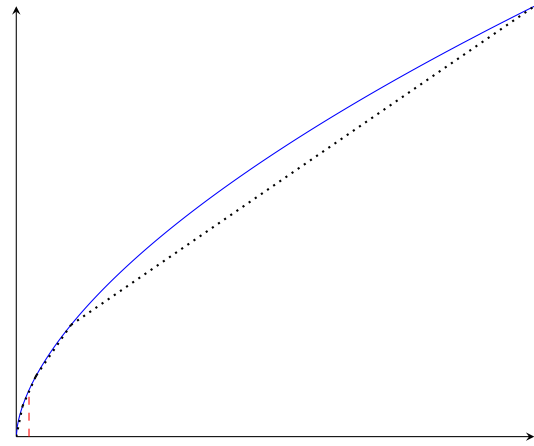
6 A^β Breakpoints

- (1, 1, 2)
 - q : 24.927400
 - $\hat{q}^\beta$: 6.886616
 - q^β : 6.886621
 - Breakpoints: [0, 2.7005882352941173, 3.460657649345137, 20.05363555217427, 24.928068483287923, 24.928068483288385, 69.05411764705883]
 - Error: 0.000004779112414
 - Relative error: 0.000000693970569
- (2, 1, 1)
 - q : 2.197860
 - $\hat{q}^\beta$: 1.550352
 - q^β : 1.603984
 - Breakpoints: [0, 1.0776943916793027, 3.3036464983028657, 3.4652942623256235, 9.439495705511714, 89.41176470588236]
 - Error: 0.053631566019673
 - Relative error: 0.033436474120665
- (2, 2, 2)
 - q : 22.020288
 - $\hat{q}^\beta$: 6.318237
 - q^β : 6.392839
 - Breakpoints: [0, 18.89081402008292, 41.49476194796521, 76.14235294117645]
 - Error: 0.074601984750085
 - Relative error: 0.011669616621065
- (3, 5, 1)
 - q : 71.165974
 - $\hat{q}^\beta$: 12.923003
 - q^β : 12.923003
 - Breakpoints: [0, 30.750913778350093, 44.57376487924321, 71.16597394333036, 71.16597394333074, 181.68117647058827]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (4, 3, 2)
 - q : 23.668856
 - $\hat{q}^\beta$: 6.675847
 - q^β : 6.675847
 - Breakpoints: [0, 19.92063386984238, 23.66863193022334, 23.668654734065058, 23.668855770059224, 181.14352941176475]
 - Error: 0.0000000000000004
 - Relative error: -0.0000000000000001

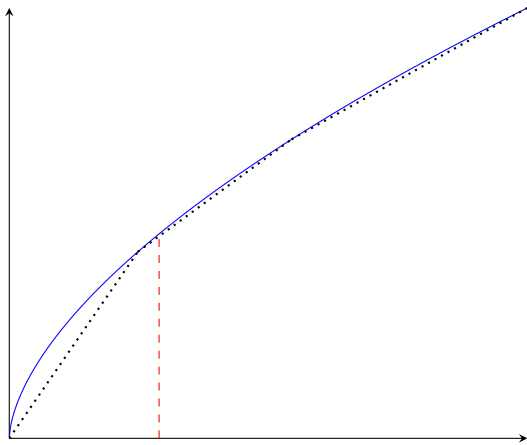
- (5, 4, 2)
 - q : 82.719688
 - $\hat{q}^\beta$: 14.143775
 - q^β : 14.143775
 - Breakpoints: [0, 11.133625686778567, 51.43476571725509, 82.7196877037192, 106.27900213658211, 192.3352941176471]
 - Error: 0.0000000000000087
 - Relative error: 0.000000000000006
- $cu, 2$
 - q : 2.696335
 - $\hat{q}^\beta$: 0.668632
 - q^β : 1.813278
 - Breakpoints: [0, 32.65629408613748]
 - Error: 1.144645709266072
 - Relative error: 0.631257870009299
- $cu, 3$
 - q : 16.738084
 - $\hat{q}^\beta$: 5.422793
 - q^β : 5.422798
 - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 13.6435791010403, 13.643579101040302, 14.376341772690381, 16.73892021814917, 16.73899708924428, 16.738997089244634, 16.738997089244645, 18.852085704732907, 128.44338870924773]
 - Error: 0.000004915147682
 - Relative error: 0.000000906385855
- $cu, 5$
 - q : 16.234975
 - $\hat{q}^\beta$: 5.324365
 - q^β : 5.324404
 - Breakpoints: [0, 6.717165864481, 9.832103793393394, 14.682070969056184, 16.23492858720698, 16.235135805353295, 20.944873037188774, 20.94487303718885, 20.944873037188852, 21.50633939847323, 22.410514202757604, 127.1036185995295]
 - Error: 0.000038309928323
 - Relative error: 0.000007195158444



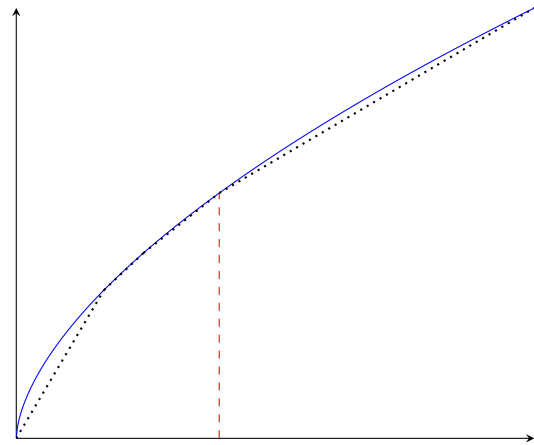
(a) $q^\beta - (1, 1, 2)$



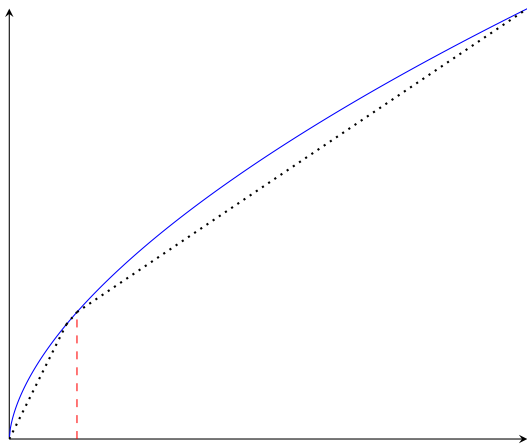
(b) $q^\beta - (2, 1, 1)$



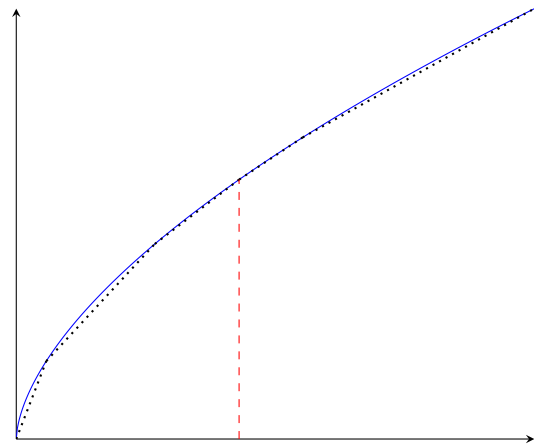
(c) $q^\beta - (2, 2, 2)$



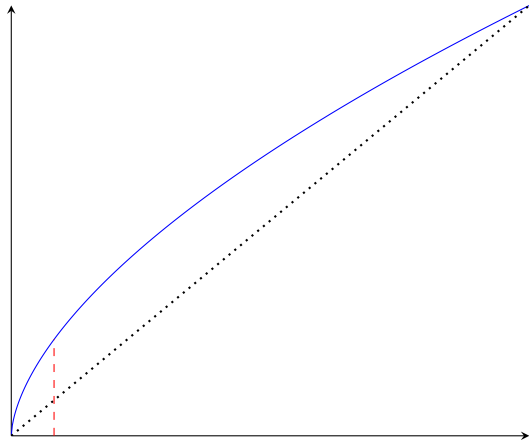
(d) $q^\beta - (3, 5, 1)$



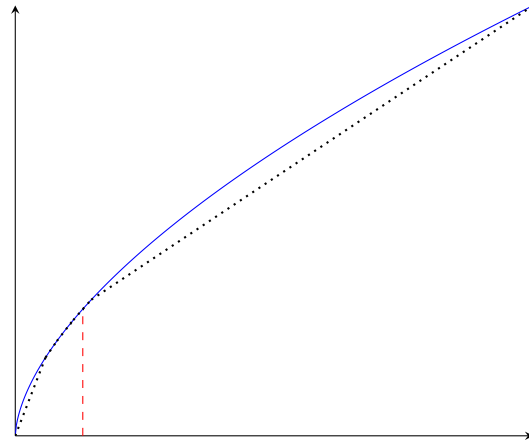
(e) $q^\beta - (4, 3, 2)$



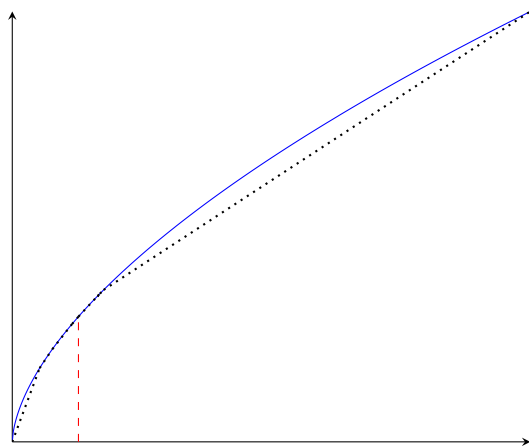
(f) $q^\beta - (5, 4, 2)$



(a) $q^\beta - cu, 2$



(b) $q^\beta - cu, 3$



(c) $q^\beta - cu, 5$

7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	-0.0	$z_{4,5,1}$	-0.0	$A_{2,5,1}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{4,5,2}$	-0.0	$A_{2,5,2}^\beta$	0.0
$z_{1,2,2}$	-0.0	$z_{5,1,1}$	-0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,1,2}$	-0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,3,2}$	-0.0	$z_{5,2,1}$	0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,2,2}$	-0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,2}$	-0.0	$z_{5,3,1}$	0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,3,2}$	-0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,2}$	-0.0	$z_{5,4,1}$	-0.0	$A_{3,4,1}^\beta$	0.0
$z_{2,1,2}$	-0.0	$z_{5,5,1}$	0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,2,1}$	0.0	$z_{5,5,2}$	-0.0	$A_{3,5,2}^\beta$	0.0
$z_{2,3,1}$	0.0	$z_{cu,1}$	0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,3,2}$	-0.0	$z_{cu,4}$	-0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,4,1}$	-0.0	$z_{hu,1}$	-0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,4,2}$	-0.0	$z_{hu,2}$	-0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,5,1}$	0.0	$z_{hu,3}$	-0.0	$A_{4,3,1}^\beta$	0.0
$z_{2,5,2}$	0.0	$z_{hu,4}$	-0.0	$A_{4,4,1}^\beta$	0.0
$z_{3,1,1}$	-0.0	$z_{hu,5}$	-0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,2}$	-0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	-0.0	$A_{1,2,1}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,1,1}^\beta$	0.0
$z_{3,3,1}$	-0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,2}$	-0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,4,1}$	-0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,2}$	-0.0	$A_{1,4,2}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,5,2}$	-0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{4,1,1}$	-0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,4,1}^\beta$	0.0
$z_{4,1,2}$	-0.0	$A_{2,1,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,2,1}$	0.0	$A_{2,2,1}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,2}$	-0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,1}^\beta$	0.0
$z_{4,3,1}$	-0.0	$A_{2,3,2}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,1}$	-0.0	$A_{2,4,1}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,4,2}$	-0.0	$A_{2,4,2}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^\beta$	0.0	$A_{4,3,1}$	0.0	$dt_{1,5,3}$	66.7
$A_{hu,4}^\beta$	0.0	$A_{4,4,1}$	0.0	$dt_{2,1,3}$	77.8
$A_{hu,5}^\beta$	0.0	$A_{4,4,2}$	0.0	$dt_{2,2,1}$	111.1
$A_{1,1,1}$	0.0	$A_{4,5,1}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,2,1}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,3,2}$	34.385977722994994
$A_{1,2,2}$	0.0	$A_{5,1,1}$	0.0	$dt_{2,3,3}$	183.3
$A_{1,3,1}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,3,2}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,4,2}$	10.0
$A_{1,4,1}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,4,3}$	55.6
$A_{1,4,2}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,5,1}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,5,2}$	68.08712401034154
$A_{1,5,2}$	0.0	$A_{5,4,1}$	0.0	$dt_{2,5,3}$	144.81792452830297
$A_{2,1,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{3,1,1}$	161.1
$A_{2,2,1}$	0.0	$A_{5,5,2}$	0.0	$dt_{3,1,2}$	10.0
$A_{2,3,1}$	0.0	$A_{cu,1}$	0.0	$dt_{3,1,3}$	10.0
$A_{2,3,2}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,4,1}$	0.0	$A_{hu,1}$	0.0	$dt_{3,2,2}$	10.0
$A_{2,4,2}$	0.0	$A_{hu,2}$	0.0	$dt_{3,2,3}$	16.14463662531543
$A_{2,5,1}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,5,2}$	0.0	$A_{hu,4}$	0.0	$dt_{3,3,2}$	10.0
$A_{3,1,1}$	0.0	$A_{hu,5}$	0.0	$dt_{3,3,3}$	188.9
$A_{3,1,2}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,1}$	144.5
$A_{3,2,1}$	0.0	$dt_{1,2,1}$	17.15006133296988	$dt_{3,4,2}$	10.0
$A_{3,2,2}$	0.0	$dt_{1,2,2}$	10.0	$dt_{3,4,3}$	10.0
$A_{3,3,1}$	0.0	$dt_{1,2,3}$	17.15006133296988	$dt_{3,5,3}$	10.0
$A_{3,3,2}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,1}$	122.2
$A_{3,4,1}$	0.0	$dt_{1,3,2}$	10.0	$dt_{4,1,2}$	144.96842105263016
$A_{3,4,2}$	0.0	$dt_{1,3,3}$	122.2	$dt_{4,1,3}$	100.0
$A_{3,5,2}$	0.0	$dt_{1,4,1}$	10.0	$dt_{4,2,1}$	122.2
$A_{4,1,1}$	0.0	$dt_{1,4,2}$	66.7	$dt_{4,2,2}$	10.0
$A_{4,1,2}$	0.0	$dt_{1,4,3}$	77.8	$dt_{4,2,3}$	106.1
$A_{4,2,1}$	0.0	$dt_{1,5,1}$	66.7	$dt_{4,3,1}$	122.2
$A_{4,2,2}$	0.0	$dt_{1,5,2}$	66.7	$dt_{4,4,1}$	122.2

Variable	Value	Variable	Value	Variable	Value
$dt_{4,4,2}$	10.0	$f_{1,5,2}^H$	0.0	$f_{5,4,1}^H$	0.0
$dt_{4,4,3}$	94.5	$f_{2,1,2}^H$	3.7711234098466413	$f_{5,5,1}^H$	17.7
$dt_{4,5,1}$	122.2	$f_{2,2,1}^H$	0.0	$f_{5,5,2}^H$	5.43555888972238
$dt_{4,5,2}$	10.0	$f_{2,3,1}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,3}$	111.1	$f_{2,3,2}^H$	0.0	$f_{1,2,1}^C$	0.0
$dt_{5,1,1}$	133.3	$f_{2,4,1}^H$	0.0	$f_{1,2,2}^C$	0.0
$dt_{5,1,2}$	138.9	$f_{2,4,2}^H$	0.0	$f_{1,3,1}^C$	0.0
$dt_{5,1,3}$	138.9	$f_{2,5,1}^H$	9.042484248424895	$f_{1,3,2}^C$	0.0
$dt_{5,2,1}$	74.66004781348396	$f_{2,5,2}^H$	0.0	$f_{1,4,1}^C$	17.3
$dt_{5,2,2}$	74.66004781348396	$f_{3,1,1}^H$	0.0	$f_{1,4,2}^C$	0.0
$dt_{5,2,3}$	10.0	$f_{3,1,2}^H$	0.0	$f_{1,5,1}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,2,1}^H$	5.214090626939785	$f_{1,5,2}^C$	13.9
$dt_{5,3,2}$	54.062951090806344	$f_{3,2,2}^H$	14.8	$f_{2,1,2}^C$	0.0
$dt_{5,3,3}$	161.1	$f_{3,3,1}^H$	0.0	$f_{2,2,1}^C$	0.0
$dt_{5,4,1}$	116.7	$f_{3,3,2}^H$	0.0	$f_{2,3,1}^C$	8.4
$dt_{5,5,1}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,3,2}^C$	0.0
$dt_{5,5,2}$	105.6	$f_{3,4,2}^H$	0.0	$f_{2,4,1}^C$	0.0
$dt_{5,5,3}$	105.6	$f_{3,5,2}^H$	0.0	$f_{2,4,2}^C$	0.0
$dt_{cu,1}$	53.3	$f_{4,1,1}^H$	12.6	$f_{2,5,1}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,1,2}^H$	0.0	$f_{2,5,2}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,2,1}^H$	0.0	$f_{3,1,1}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,2,2}^H$	0.0	$f_{3,1,2}^C$	0.9981482756472416
$dt_{hu,3}$	18.9	$f_{4,3,1}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,1}^H$	0.0	$f_{3,2,2}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,4,2}^H$	0.0	$f_{3,3,1}^C$	0.0
$f_{1,1,1}^H$	8.8	$f_{4,5,1}^H$	0.0	$f_{3,3,2}^C$	0.0
$f_{1,2,1}^H$	0.0	$f_{4,5,2}^H$	0.0	$f_{3,4,1}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,1,1}^H$	0.0	$f_{3,4,2}^C$	0.0
$f_{1,3,1}^H$	0.0	$f_{5,1,2}^H$	0.0	$f_{3,5,2}^C$	0.0
$f_{1,3,2}^H$	0.0	$f_{5,2,1}^H$	0.0	$f_{4,1,1}^C$	0.0
$f_{1,4,1}^H$	0.0	$f_{5,2,2}^H$	0.0	$f_{4,1,2}^C$	0.0
$f_{1,4,2}^H$	0.0	$f_{5,3,1}^H$	0.0	$f_{4,2,1}^C$	6.1
$f_{1,5,1}^H$	0.0	$f_{5,3,2}^H$	0.0	$f_{4,2,2}^C$	0.0

Variable	Value	Variable	Value	Variable	Value
$f_{4,3,1}^C$	0.0	$\text{RecLMTD}_{3,2,1}$	0.1	$q_{1,2,1}$	0.0
$f_{4,4,1}^C$	0.0	$\text{RecLMTD}_{3,2,2}$	0.1	$q_{1,2,2}$	0.0
$f_{4,4,2}^C$	0.0	$\text{RecLMTD}_{3,3,1}$	0.1	$q_{1,3,1}$	0.0
$f_{4,5,1}^C$	0.0	$\text{RecLMTD}_{3,3,2}$	0.026450800778310225	$q_{1,3,2}$	0.0
$f_{4,5,2}^C$	0.0	$\text{RecLMTD}_{3,4,1}$	0.1	$q_{1,4,1}$	0.0
$f_{5,1,1}^C$	0.0	$\text{RecLMTD}_{3,4,2}$	0.1	$q_{1,4,2}$	0.0
$f_{5,1,2}^C$	0.0	$\text{RecLMTD}_{3,5,2}$	0.1	$q_{1,5,1}$	0.0
$f_{5,2,1}^C$	0.0	$\text{RecLMTD}_{4,1,1}$	0.00926241094037416	$q_{1,5,2}$	0.0
$f_{5,2,2}^C$	0.0	$\text{RecLMTD}_{4,1,2}$	0.01017066110438082	$q_{2,1,2}$	0.0
$f_{5,3,1}^C$	0.0	$\text{RecLMTD}_{4,2,1}$	0.1	$q_{2,2,1}$	0.0
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,2,2}$	0.02307928660940034	$q_{2,3,1}$	0.0
$f_{5,4,1}^C$	0.0	$\text{RecLMTD}_{4,3,1}$	0.1	$q_{2,3,2}$	0.0
$f_{5,5,1}^C$	0.0	$\text{RecLMTD}_{4,4,1}$	0.02230651890829312	$q_{2,4,1}$	0.0
$f_{5,5,2}^C$	0.0	$\text{RecLMTD}_{4,4,2}$	0.024030191698075007	$q_{2,4,2}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,5,1}$	0.1	$q_{2,5,1}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{4,5,2}$	0.02554670259920889	$q_{2,5,2}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.1	$\text{RecLMTD}_{5,1,1}$	0.007344552824542799	$q_{3,1,1}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,1,2}$	0.007199424046076313	$q_{3,1,2}$	0.0
$\text{RecLMTD}_{1,3,2}$	0.1	$\text{RecLMTD}_{5,2,1}$	0.1	$q_{3,2,1}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,2,2}$	0.1	$q_{3,2,2}$	0.0
$\text{RecLMTD}_{1,4,2}$	0.01377039538464575	$\text{RecLMTD}_{5,3,1}$	0.1	$q_{3,3,1}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,3,2}$	0.028888218232147534	$q_{3,3,2}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.1	$\text{RecLMTD}_{5,4,1}$	0.014265344121837218	$q_{3,4,1}$	0.0
$\text{RecLMTD}_{2,1,2}$	0.1	$\text{RecLMTD}_{5,5,1}$	0.1	$q_{3,4,2}$	0.0
$\text{RecLMTD}_{2,2,1}$	0.1	$\text{RecLMTD}_{5,5,2}$	0.0094696969696968	$q_{3,5,2}$	0.0
$\text{RecLMTD}_{2,3,1}$	0.046808918968723766	$\text{RecLMTD}_{cu,1}$	0.016108680951407914	$q_{4,1,1}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.016606920715186237	$\text{RecLMTD}_{cu,4}$	0.008535688158765427	$q_{4,1,2}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.1	$\text{RecLMTD}_{hu,1}$	0.012387051800747543	$q_{4,2,1}$	0.0
$\text{RecLMTD}_{2,4,2}$	0.1	$\text{RecLMTD}_{hu,2}$	0.045618305265862594	$q_{4,2,2}$	0.0
$\text{RecLMTD}_{2,5,1}$	0.1	$\text{RecLMTD}_{hu,3}$	0.04305097282506776	$q_{4,3,1}$	0.0
$\text{RecLMTD}_{2,5,2}$	0.03361831879010532	$\text{RecLMTD}_{hu,4}$	0.015434474530129737	$q_{4,4,1}$	0.0
$\text{RecLMTD}_{3,1,1}$	0.1	$\text{RecLMTD}_{hu,5}$	0.024141568940966463	$q_{4,4,2}$	0.0
$\text{RecLMTD}_{3,1,2}$	0.1	$q_{1,1,1}$	0.0	$q_{4,5,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{4,5,2}$	0.0	$t_{2,5,2}^H$	142.9043734441421	$t_{1,4,1}^C$	176.7
$q_{5,1,1}$	0.0	$t_{3,1,1}^H$	226.7	$t_{1,4,2}^C$	129.3704699676358
$q_{5,1,2}$	0.0	$t_{3,1,2}^H$	122.35608108108104	$t_{1,5,1}^C$	148.85529774887698
$q_{5,2,1}$	0.0	$t_{3,2,1}^H$	226.7	$t_{1,5,2}^C$	93.3
$q_{5,2,2}$	0.0	$t_{3,2,2}^H$	122.35608108108063	$t_{2,1,2}^C$	60.0
$q_{5,3,1}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,2,1}^C$	221.7
$q_{5,3,2}$	0.0	$t_{3,3,2}^H$	74.52388659510075	$t_{2,3,1}^C$	221.1
$q_{5,4,1}$	0.0	$t_{3,4,1}^H$	226.7	$t_{2,3,2}^C$	37.8
$q_{5,5,1}$	0.0	$t_{3,4,2}^H$	65.6	$t_{2,4,1}^C$	82.2
$q_{5,5,2}$	0.0	$t_{3,5,2}^H$	65.6	$t_{2,4,2}^C$	126.81811071033897
$q_{cu,1}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,5,1}^C$	167.42543850429928
$q_{cu,4}$	0.0	$t_{4,1,2}^H$	211.55392800990342	$t_{2,5,2}^C$	93.3
$q_{hu,1}$	0.0	$t_{4,2,1}^H$	204.06666666666666	$t_{3,1,1}^C$	127.3787644042812
$q_{hu,2}$	0.0	$t_{4,2,2}^H$	211.55392800990342	$t_{3,1,2}^C$	60.0
$q_{hu,3}$	0.0	$t_{4,3,1}^H$	271.1	$t_{3,2,1}^C$	221.7
$q_{hu,4}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,2,2}^C$	115.6
$q_{hu,5}$	0.0	$t_{4,4,2}^H$	211.55392800990342	$t_{3,3,1}^C$	221.1
$t_{1,1,1}^H$	160.0	$t_{4,5,1}^H$	271.1	$t_{3,3,2}^C$	37.8
$t_{1,2,1}^H$	160.0	$t_{4,5,2}^H$	211.55392800990342	$t_{3,4,1}^C$	176.7
$t_{1,2,2}^H$	160.0	$t_{5,1,1}^H$	179.34745762711862	$t_{3,4,2}^C$	82.2
$t_{1,3,1}^H$	160.0	$t_{5,1,2}^H$	143.24166839309086	$t_{3,5,2}^C$	166.23812949640183
$t_{1,3,2}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,1}^C$	60.00012362758214
$t_{1,4,1}^H$	160.0	$t_{5,2,2}^H$	108.74827200025184	$t_{4,1,2}^C$	60.0
$t_{1,4,2}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,2,1}^C$	221.7
$t_{1,5,1}^H$	160.0	$t_{5,3,2}^H$	106.53559322033877	$t_{4,2,2}^C$	115.6
$t_{1,5,2}^H$	160.0	$t_{5,4,1}^H$	198.9	$t_{4,3,1}^C$	221.1
$t_{2,1,2}^H$	233.6062112046282	$t_{5,5,1}^H$	198.9	$t_{4,4,1}^C$	176.7
$t_{2,2,1}^H$	139.85985526856254	$t_{5,5,2}^H$	198.9	$t_{4,4,2}^C$	82.2
$t_{2,3,1}^H$	248.9	$t_{1,1,1}^C$	90.20394736842229	$t_{4,5,1}^C$	173.69197447977496
$t_{2,3,2}^H$	137.8	$t_{1,2,1}^C$	221.7	$t_{4,5,2}^C$	93.3
$t_{2,4,1}^H$	248.9	$t_{1,2,2}^C$	115.6	$t_{5,1,1}^C$	92.62135922330096
$t_{2,4,2}^H$	233.6062112046282	$t_{1,3,1}^C$	221.1	$t_{5,1,2}^C$	60.0
$t_{2,5,1}^H$	248.9	$t_{1,3,2}^C$	215.35238095238094	$t_{5,2,1}^C$	221.7

Variable	Value	Variable	Value	Variable	Value
$t_{5,2,2}^C$	115.6	$b_{4,1,2}^{H,in}$	0.0	$b_{2,5,2}^{H,out}$	0.0
$t_{5,3,1}^C$	221.1	$b_{4,2,1}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	0.0
$t_{5,3,2}^C$	37.8	$b_{4,2,2}^{H,in}$	0.0	$b_{3,1,2}^{H,out}$	0.0
$t_{5,4,1}^C$	176.7	$b_{4,3,1}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	1182.0343451272493
$t_{5,5,1}^C$	99.59775971067485	$b_{4,4,1}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	1810.8699999999995
$t_{5,5,2}^C$	124.01079136690646	$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	1408.0	$b_{4,5,1}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	0.0
$b_{1,2,1}^{H,in}$	0.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,2,2}^{H,in}$	0.0	$b_{5,1,1}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	0.0
$b_{1,3,1}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	0.0	$b_{3,5,2}^{H,out}$	0.0
$b_{1,3,2}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	0.0	$b_{4,1,1}^{H,out}$	3415.86
$b_{1,4,1}^{H,in}$	0.0	$b_{5,2,2}^{H,in}$	0.0	$b_{4,1,2}^{H,out}$	0.0
$b_{1,4,2}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	0.0	$b_{4,2,1}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	0.0	$b_{4,2,2}^{H,out}$	0.0
$b_{1,5,2}^{H,in}$	0.0	$b_{5,4,1}^{H,in}$	0.0	$b_{4,3,1}^{H,out}$	0.0
$b_{2,1,2}^{H,in}$	877.070805876874	$b_{5,5,1}^{H,in}$	3520.5300000000007	$b_{4,4,1}^{H,out}$	0.0
$b_{2,2,1}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	1081.1326631657814	$b_{4,4,2}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	1408.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	0.0	$b_{1,2,1}^{H,out}$	0.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,4,1}^{H,in}$	0.0	$b_{1,2,2}^{H,out}$	0.0	$b_{5,1,1}^{H,out}$	0.0
$b_{2,4,2}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	0.0
$b_{2,5,1}^{H,in}$	2250.6743294329563	$b_{1,3,2}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	0.0
$b_{2,5,2}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	0.0
$b_{3,1,1}^{H,in}$	0.0	$b_{1,4,2}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	0.0
$b_{3,1,2}^{H,in}$	0.0	$b_{1,5,1}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	0.0
$b_{3,2,1}^{H,in}$	1182.0343451272493	$b_{1,5,2}^{H,out}$	0.0	$b_{5,4,1}^{H,out}$	0.0
$b_{3,2,2}^{H,in}$	1810.8699999999995	$b_{2,1,2}^{H,out}$	877.070805876874	$b_{5,5,1}^{H,out}$	3520.5300000000007
$b_{3,3,1}^{H,in}$	0.0	$b_{2,2,1}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	1081.1326631657814
$b_{3,3,2}^{H,in}$	0.0	$b_{2,3,1}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	0.0	$b_{1,2,1}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	0.0	$b_{2,4,1}^{H,out}$	0.0	$b_{1,2,2}^{C,in}$	0.0
$b_{3,5,2}^{H,in}$	0.0	$b_{2,4,2}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	3415.86	$b_{2,5,1}^{H,out}$	2250.6743294329563	$b_{1,3,2}^{C,in}$	0.0

Variable	Value	Variable	Value	Variable	Value
$b_{1,4,1}^{C,in}$	3056.91	$b_{5,2,2}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	0.0
$b_{1,4,2}^{C,in}$	0.0	$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	1352.37
$b_{1,5,1}^{C,in}$	0.0	$b_{5,3,2}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0
$b_{1,5,2}^{C,in}$	1296.87	$b_{5,4,1}^{C,in}$	0.0	$b_{4,3,1}^{C,out}$	0.0
$b_{2,1,2}^{C,in}$	0.0	$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	0.0
$b_{2,2,1}^{C,in}$	0.0	$b_{5,5,2}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0
$b_{2,3,1}^{C,in}$	1857.24	$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,1}^{C,out}$	0.0
$b_{2,3,2}^{C,in}$	0.0	$b_{1,2,1}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	0.0
$b_{2,4,1}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,1,1}^{C,out}$	0.0
$b_{2,4,2}^{C,in}$	0.0	$b_{1,3,1}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{2,5,2}^{C,in}$	0.0	$b_{1,4,1}^{C,out}$	3056.91	$b_{5,2,2}^{C,out}$	0.0
$b_{3,1,1}^{C,in}$	0.0	$b_{1,4,2}^{C,out}$	0.0	$b_{5,3,1}^{C,out}$	0.0
$b_{3,1,2}^{C,in}$	59.888896538834494	$b_{1,5,1}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	1296.87	$b_{5,4,1}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{2,1,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	0.0
$b_{3,3,1}^{C,in}$	0.0	$b_{2,2,1}^{C,out}$	0.0	$b_{5,5,2}^{C,out}$	0.0
$b_{3,3,2}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	1857.24		
$b_{3,4,1}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	0.0		
$b_{3,4,2}^{C,in}$	0.0	$b_{2,4,1}^{C,out}$	0.0		
$b_{3,5,2}^{C,in}$	0.0	$b_{2,4,2}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	0.0	$b_{2,5,2}^{C,out}$	0.0		
$b_{4,2,1}^{C,in}$	1352.37	$b_{3,1,1}^{C,out}$	0.0		
$b_{4,2,2}^{C,in}$	0.0	$b_{3,1,2}^{C,out}$	59.8888965388345		
$b_{4,3,1}^{C,in}$	0.0	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	0.0	$b_{3,4,1}^{C,out}$	0.0		
$b_{5,1,1}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	0.0		
$b_{5,1,2}^{C,in}$	0.0	$b_{3,5,2}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0		